

Boundary Spanning Individuals: Their Role in Information Transfer and Their Antecedents¹

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An investigation of alternative mechanisms by which information is imported into organizations indicates that informational boundary spanning is accomplished only by those individuals who are well connected internally and externally. These key individuals are nominated as technically competent in their unit and have personal characteristics to link effectively their unit to external areas.

A member of the laboratory's technical staff needs up-to-date marketing information. What does she do? She knows that the formal report system is not current and her last conversation with a marketing specialist was not productive.

A project leader in the development division needs to know the latest developments in fiberoptics for an upcoming report. Further, he has a journal article in a related technical area that he does not know how to interpret. Should he invest more time and money exploring this technology? Where will this project manager get this information and expertise?

Individuals in organizations are continuously called upon to make decisions without sufficient information. Given the unrelenting pace of organizational life, decision makers must be able to maximize their information acquisition at a relatively low cost in terms of time, energy, and effort (Arrow, 1974; Mintzberg, 1973). More generally, organizations must acquire timely information from a number of external areas; further, this information must be translated so that it is of relevance to the organization. How, then, are these individuals in particular and organizations more generally effectively linked to external areas?

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Information acquired via formal information media (e.g., MIS reports) is inherently dated, and formal information systems have only limited encoding capabilities (Gerstberger & Allen, 1968). Widespread direct communication across organizational boundaries is costly, inefficient, and prone to bias (Katz & Kahn, 1966; Wilensky, 1967; Allen, 1977). Informal social mechanisms, however, are an effective medium for acquiring and encoding timely, current, and soft information (March & Simon, 1958; Edstrom & Galbraith, 1977; Aldrich & Herker, 1977). The literature, however, is equivocal as to the nature of these information mechanisms linking organizations with external areas. Some research suggests that individuals with strong links to external areas provide external linkage (Leifer & Huber, 1977; Bacharach & Aiken, 1977; Miles, 1977; Keller & Holland, 1975). Others argue that a multistep process is required to link organizations to external areas (Pettigrew, 1972; Allen & Cohen, 1969; Schwartz & Jacobson, 1977; Tushman, 1977).

The present research distinguishes between external communication stars—those individuals with substantial boundary spanning activity (BSA)—internal communication stars, and those individuals who are both internal and external communication stars (boundary spanning individuals). It tests the extent to which each communication role links its unit to external sources of information. This research also investigates the antecedents of boundary role status, building on research that suggests that boundary spanning individuals are seen by their colleagues as technically competent and have the background and skills to communicate with different external areas (Blau, 1963; Tushman & Scanlan, 1981).

COMMUNICATION BOUNDARIES AND BOUNDARY SPANNING

As organizations evolve, specialized units are created to deal with relatively homogeneous tasks and with specific segments of the organization's task environment (for instance, marketing vs. R&D departments). Individuals with specialized skills and training are hired and socialized to attend to the unit's particular work requirements. Organizational boundaries, then, separate specialized subunits from each other and from external areas (Katz & Kahn, 1966).

Specialization and the existence of organizational boundaries are also associated with the evolution of local norms, values, and languages tailored to the requirements of the unit's work (March & Simon, 1958; Lawrence & Lorsch, 1967; Ritti & Gouldner, 1969). This specialization is a double-edged sword, for it increases the efficiency of information processing within the unit, but simultaneously creates obstacles to information processing between the unit and external areas. More specifically, organizational boundaries are associated with a communication boundary or impedance separating the unit from external areas. Communication boundaries are created by the interaction between (1) idiosyncratic language/coding schemes and (2) by the development of local conceptual frameworks.

Special technical languages evolve in units as members attend to their local work requirements (Arrow, 1974; March & Simon, 1958). Actors evolve a shared semantic space, which permits them to express large amounts of information with relatively few symbols (Randolph, 1978; Cherry, 1965; Allen, 1977). This specialization of both verbal and nonverbal language permits an economical transmission of information and minimizes misunderstandings between actors who share the language.

Because units have their own specialized semantics, it becomes necessary to recode at boundaries between units. The objective in this translation is to convert words into the second semantic space while retaining meanings held in the first. A loss of information, or semantic noise, is almost inevitable at each translation because words take on a richness of meaning that extends beyond simple associations to the full set of rules, habits, and conventions for the word's use (Bar-Hillel & Carnap, 1953; Parkinson, 1968). As an example, note the difficulty in communicating the meaning of an idiomatic expression.

Members of subunits also develop their own local social constructions of reality to help them define and interpret their social world (Berger & Luckman, 1966; Van Maanen & Katz, 1979). These collective beliefs provide rules (e.g., premises of decision making, levels of satisficing) by which actors can define and manage the unique and recurring problems of organizational life. These shared beliefs provide the context and rules with which information is processed. The meaning of an event or an expression, then, can only be inferred when both its context and content have been both enacted and encoded (Bateson, 1972; Watzlawick, Beavin, & Jackson, 1967). For example, note the contextual clues needed to encode accurately an ironic message.

Communication, then, is inherently ambiguous. The information content of a communication episode can only be ascertained by first registering and then encoding the equivocality of the message (Cherry, 1965; Weick, 1969). The interaction of local languages and local conceptual schemes make consistent enactment and encoding problematic. Communication across boundaries, therefore, is difficult and prone to bias and distortion. The greater the language/cognitive differences, the greater the communication impedance (March & Simon, 1958; Miles, 1977; Katz & Kahn, 1966; Simon, 1979; Dearborn & Simon, 1958).

Boundary Spanning as a Two-Step Process

Someone learning a new language must learn both the vocabulary and how to interpret contextual messages—that is nonverbal signs, intonations of voice, the use of space (Watzlawick et al., 1967). Similarly, communicating across organizational boundaries requires learning the local coding schemes and languages as well as the specialized conceptual frameworks. Boundaries can be spanned effectively only by individuals who understand the coding schemes are attuned to the contextual information on both

sides of the boundary, enabling them to search out relevant information on one side and disseminate it on the other. Informational boundary spanning, then, must be a two part process; obtaining information from outside units and disseminating this information to internal users.

Much of the literature on boundary spanning identifies and investigates only one component of this two part process. The research on BSA focuses on those individuals who have strong linkages outside the organization (Keller & Holland, 1975; Leifer & Huber, 1977; Adams, 1976; Miles, 1977; Bacharach & Aiken, 1977). Although these individuals may represent the organization externally (e.g., salesmen), there is no evidence that these external communication stars actually import information (Von Hippel, 1976). What is frequently referred to as boundary spanning may be simply external activity with little information disseminated within the organization; indeed, these individuals are frequently internal isolates (Roberts & O'Reilly, 1979; Allen, 1970).

Other research has focused on internal communication stars (Whitley & Frost, 1973; Frost & Whitley, 1971; Allen & Cohen, 1969; Baldrige & Burnham, 1975; Tushman, 1977). This research has inferred that internal stars also link their colleagues to external areas. These individuals may or may not be strongly linked externally. Indeed, research by Tushman and Scanlan (1981) found that some internal stars were locally oriented (not strongly linked externally) and others were more cosmopolitan.

Internal linkage and external linkage are necessary but not sufficient conditions to perform an informational boundary spanning function. Informational boundary spanning will be accomplished *only* by those individuals who are well connected to external information areas and who also are well connected internally and thus able to disseminate new information and new ideas to their more locally oriented colleagues. These ideas suggest several hypotheses:

Hypothesis 1: There will be a positive association between being chosen as a valuable source of new information and ideas and being a boundary spanning individual.

Hypothesis 1A: There will be no association between being an internal or external star and being chosen as a valuable source of new information and ideas.

Is informational boundary spanning more than a function of formal status? Formal position facilitates both internal and external communication, but it may or may not be associated with boundary role status (Katz & Tushman, 1979). Research from several disciplines suggests that perceived work-related competence is a more basic determinant of boundary role status than formal position. Technical competence, partly a function of information access internally and externally, stimulates internal consultation and may lead to a promotion. Thus although there will be overlap between formal position and boundary role status, informational boundary spanning is not a function of formal position (Blau, 1963; Tushman & Scanlan, 1981).

Hypothesis 2: For nonmanagerial personnel, there will be a positive association between being chosen as a valuable source of new information and ideas and being a boundary spanning individual.

What are the antecedents of informational boundary spanning? Boundary spanning individuals must be well connected internally and externally. Previous research indicates that perceived work-related competence is a primary determinant of internal consultation; people will approach those whom they see as competent, independent of formal status (Spekman, 1979; Blau, 1963). Perceived technical competence may lead to a promotion, which in turn would facilitate both internal and external linkages. These ideas indicate that both formal status and perceived technical competence are primary determinants of internal consultation, but perceived competence is the more basic determinant (Tushman & Scanlan, 1981).

Hypothesis 3: There will be a positive association between being an internal communication star and being nominated as technically competent. This association will not exist for external stars.

Hypothesis 3A: Perceived technical competence will be a stronger determinant of internal star status than formal position in the hierarchy.

Boundary spanning individuals also must be able to communicate effectively with external areas; they must be able to understand the languages and conceptual frameworks of individuals outside their unit (for example, areas in the larger organization or areas outside the organization). To communicate effectively with external areas, external communication stars must have the background, experience, and training to deal with the communication impedance separating his or her unit from external areas. Because subunits face several relevant external areas, external stars will have different characteristics to meet the information requirements of each distinct area (Tushman, 1977; Katz & Tushman, 1979).

Hypothesis 4: External stars will have specialized characteristics to match information requirements of distinct external areas.

Informational boundary spanning, then, is a two-part process: obtaining information from external areas and disseminating that information to internal users. The ability of an individual to span a boundary is predicated on his/her having the work-related expertise required to communicate effectively on both sides of the communication boundary.

METHOD

This study was conducted in the research and development (R&D) division of an American high-technology medical instrument corporation. This laboratory was divided into four departments and employed 210 professionals. Secretarial and nontechnical personnel were excluded from this study.

Technical Communication

To provide communication data, individuals were asked to report their actual, work-related, oral communication for a number of selected days (respondents were asked not to report strictly social communication, nor were they to report written communication). These data were collected once a week for five weeks (an equal number of each of the weekdays was chosen). The overall response rate was 74 percent. Moreover, 81 percent of all communication reported within the laboratory was mentioned by both parties. See Tushman (1977) or Weiss and Jacobson (1960) for comparative data. Extra-laboratory data were self-report data that could not be corroborated with discussion partners.

The department was chosen as the unit of analysis, and three distinct communication roles were identified: internal stars, external stars, and boundary spanning individuals. Internal communication stars, identified for each department, were defined as those individuals who engaged in more than one standard deviation above the mean number of intra-departmental communications (as reported by their peers). Three external areas were identified: one within the organization and two outside. The intra-organizational area included other areas within the corporation (for instance, marketing or manufacturing). The two extra-organizational areas were (1) professional areas (universities and/or professional societies) and (2) operational areas (suppliers, vendors, or customers). External stars for each area were defined as those individuals in the top fifth of the extra-department communication distribution. (The extra-departmental communication distributions were highly skewed. Many individuals had no communication with external areas, and a few had substantial external communication. Thus a one standard deviation rule could not be used. The top fifth criterion provided for approximately equal numbers of external stars for each area.) Finally, boundary spanning individuals were defined as those individuals who were both internal and external communication stars.

Demographic Data and Peer Nominations

Approximately 75 percent of the professional staff responded to a questionnaire which asked (among other things): tenure (number of years in the division); education (highest degree earned); papers (number of papers published or presented in the past five years); hours reading (average number of hours per week spent reading professional literature); and status (position in the divisional chain of command). Status has an inverted scale, ranging from 1 = vice president to 6 = entry level employee. The ranges for the other variables were: tenure (0-24 years), education (1 = did not complete high school to 6 = Ph.D.), papers (0-15 papers), and hours reading (0-12 hours).

To gather data on perceived technical competence, each individual was asked to nominate those "who are the most technically competent people in your department." The distribution of nominations per individual was highly skewed (two thirds of the individuals had one or fewer nominations). Individuals were categorized as technically competent if nominated by two or more individuals ($N=75$).

Individuals also were asked to nominate those in their department "whom you consider to be valuable contacts for external information, new ideas, and critical evaluation." This distribution also was highly skewed; individuals were categorized as valuable sources of new ideas and information if they were nominated by two or more individuals ($N=91$).

Finally, each individual was asked to indicate how he/she developed contacts with each of the four external areas. The categories, based on Granovetter's (1973) work, were: formal channels, previously worked together, attended the same school, were neighbors, via mutual acquaintance, and others. The last four categories were combined as they were chosen infrequently.

RESULTS

Hypothesis 1 is based on the idea that informational boundary spanning is a two-part process; obtaining information from external sources and disseminating information to internal users. It was predicted that there would be a positive association between being a boundary spanning individual and being chosen as a valuable source of new information and ideas.

In support of Hypothesis 1, there is a positive association between being nominated as a valuable source of external information and new ideas and being a boundary spanning individual. This positive association, however, does not exist for either internal stars or external stars (Table 1). Furthermore, the absolute number of nominations as sources of new information and ideas for each communication role are significantly different (Kruskal Wallis ANOVA; $p<.001$). Boundary spanning individuals have about twice the number of nominations as do internal stars and almost three times the number of nominations as do external stars. There then is strong support for Hypothesis 1. Those who are nominated as valuable sources of new information and ideas are *both* internal and external stars. Neither internal stars nor external stars are nominated as valuable sources of new information by their peers.

The second hypothesis reasons that boundary spanning is not simply a function of formal status; that nonmanagerial boundary spanning individuals would be seen by their colleagues as more valuable sources of new information and ideas than would nonmanagerial internal or external stars. Of the boundary spanning individuals, 32 percent (7 out of 22) were also supervisors. After eliminating these individuals, there still is a strong association between boundary spanning position and being nominated as a

TABLE 1
Individuals Nominated as
"Valuable Sources of External Information
and New Ideas" and Communication Roles

A. Boundary Spanners

	<i>Boundary Spanners</i>	<i>Nonspanners</i>	<i>Total</i>
Valuable sources	19	72	91
Unnamed	3	110	113
Total	22	182	204

$\chi^2 = 15.558$ $p < .001$
 Contingency Coefficient = .27^a

B. Internal Stars—excluding Boundary Spanners

	<i>Stars</i>	<i>Nonstars</i>	<i>Total</i>
Valuable sources	8	64	72
Unnamed	7	103	110
Total	15	167	182

$\chi^2 = 1.342$ $p = \text{N.S.}$
 Contingency Coefficient = .08

C. External Stars—excluding Boundary Spanners

	<i>Stars</i>	<i>Nonstars</i>	<i>Total</i>
Valuable sources	31	41	72
Unnamed	36	74	110
Total	67	115	182

$\chi^2 = 2.000$ $p = \text{N.S.}$
 Contingency Coefficient = .10

^aUpper limit for 2 × 2 contingency coefficient is .71.

valuable source of new information and ideas (Table 2). Similarly, non-managerial boundary spanners are nominated as valuable sources of new information significantly more often than are nonmanagerial internal and external stars (Kruskal Wallis ANOVA; $p < .05$). Although formal status is helpful (all 7 of the supervisory boundary spanners are nominated as valuable sources of information), boundary spanning is more than a function of formal position in the hierarchy.

What are the relative contributions of being a boundary spanning individual, an internal or external star, or formal status in predicting nominations as a valuable source of new information and ideas? Consistent with the results shown above, a simultaneous multiple regression indicates that only being a boundary spanning individual and formal status are important determinants of being nominated. (The dependent variable was categorized as 0, 1, or 2 because of the distribution's extreme degree of skew.) Boundary spanning status is of greater importance than formal status in predicting the number of nominations as a valuable source of new information and ideas. The regression equation (with standardized betas) is:

TABLE 2
Individuals Nominated as "Valuable
Sources of External Information and New Ideas"
and Communication Roles for Nonmanagers

	<i>Boundary Spanners</i>	<i>Nonspanners</i>	<i>Total</i>
Valuable sources	12	60	72
Unnamed	3	105	108
Total	15	165	180

$\chi^2 = 9.167$ ($p < .003$)
Contingency Coefficient = .23

Number of nominations =

$$-.20^* (\text{status}) + .31^{**} (\text{BSI}) + .08 (\text{internal star}) - .04 (\text{external star})$$

$$F = 8.77; df = 4, 198 \quad R^2 = .39$$

$$^*p < .05$$

$$^{**}p < .01$$

What are the antecedents of those individuals who are internal and external stars? Hypothesis 3 suggests that there would be a positive association between being an internal star and being nominated as technically competent, and that perceived technical competence would be a more important determinant of internal star status than formal position in the hierarchy.

In support of Hypothesis 3, there is a significant positive association between being an internal communication star and being nominated as technically competent by one's peers. This association holds even after boundary spanning individuals are eliminated (Table 3). Individuals, then, approach those peers who are seen as competent, independent of their external linkage. Although internal stars are seen as technically competent, there is no association between being an external star and perceived competence.

To address Hypothesis 3A, a simultaneous discriminant analysis was employed to investigate the relative contributions of formal status and perceived technical competence in predicting internal stars. Both factors predict internal star status, but perceived competence is a more powerful predictor than formal status (the standardized discriminant coefficients are .74 and .52, respectively, the discriminant function being significant at less than the .001 level).

Hypothesis 4 suggests that external communication stars must have specialized characteristics and backgrounds to deal with the unique information requirements of different external areas. This broad hypothesis can be made more specific to the four external areas studied here.

TABLE 3
Individuals Nominated as Technically
Competent and Communication Roles

<i>A. Internal Stars—Including Boundary Spanners</i>			
	<i>Stars</i>	<i>Nonstars</i>	<i>Total</i>
Competent	23	52	75
Unnamed	14	115	129
Total	37	167	204
$X^2 = 11.241 \ p < .001$ Contingency Coefficient = .23			
<i>B. Internal Stars—Excluding Boundary Spanners</i>			
	<i>Stars</i>	<i>Nonstars</i>	<i>Total</i>
Competent	10	52	62
Unnamed	5	115	120
Total	15	167	182
$X^2 = 7.512 \ p < .01$ Contingency Coefficient = .20			
<i>C. External Stars—Excluding Boundary Spanners</i>			
	<i>Stars</i>	<i>Nonstars</i>	<i>Total</i>
Competent	23	39	62
Unnamed	44	76	120
Total	67	115	182
$X^2 = .01 \ p < N.S.$ Contingency Coefficient = .01			

Effective communication with other laboratory areas requires that the individual be familiar with the different work requirements and the locus of expertise in those areas. Literature on professional communication indicates that informal mechanisms are more vital than formal mechanisms for developing contacts and that having worked together is a major determinant of professional communication (Allen, 1977). Although working together may provide an informal linkage, formal status will also facilitate intra-laboratory communication. Communication to the corporation, on the other hand, is more constrained to formal mechanisms. Communication regarding schedules, costs, and budgets can best be accomplished via the formal hierarchy (Rosenbloom & Wolek, 1970).

Effective linkage to professional areas outside the organization can be accomplished only by those individuals with technical training and cosmopolitan orientation, that is, by those who are more professional (Kornhauser, 1963; Hage, 1974). Literature on professional communication also indicates that effective linkages are forged more through informal methods (e.g., worked together, contact at professional meetings) than through formal methods (Crane, 1972; Kronhauser, 1962). Communication with suppliers, vendors, or customers is more organizationally focused. A basic

determinant of operational communication, then, is formal status or formal contact via prescribed channels (Tushman, 1979; Whitley & Frost, 1973).

Several analyses were conducted to test these specific hypotheses regarding the characteristics of external communication stars. Table 4 presents the results of several discriminant analyses of the characteristics of stars versus nonstars for each external area. Table 5 presents data on the methods used by communication stars to forge links with each area.

Contacts within the laboratory are established more through working together than through formal channels (Table 5). Neither status nor professional characteristics, however, serve to distinguish between stars and nonstars within the laboratory (Table 4). For corporate communication, on the other hand, contacts are forged more frequently through formal channels, and formal status is an important determinant of corporate communication. Organizational communication, then, is based on formal mechanisms, and laboratory communication is based more heavily on informal mechanisms.

Those individuals who are strongly linked to professional areas are more professionally oriented (in terms of papers presented, hours spent reading the literature) than are those individuals weakly linked to professional areas (Table 4). Professional communication bonds are forged through both formal and informal mechanisms (worked together, mutual acquaintances), and links to operational areas are developed primarily through formal channels. No demographic characteristics discriminate between operational stars and nonstars. As a set, then, these results indicate that communication stars in different areas develop their contacts differently, and they have specialized characteristics that are consistent with the information requirements of their particular communication boundary.

TABLE 4
Discriminant Analysis of Stars versus Nonstars
for Four External Areas

<i>Domain</i>	<i>Variables</i>	<i>Discriminant Coefficients</i>	<i>Variable F Score</i>	<i>Overall F Score</i>	<i>Percent Classified</i>
Laboratory:	Status	.608	1.589	1.345 (4,127 <i>DF</i>)	62.8
	Education	.925	1.504		
	Hours read	.495	1.206		
	Papers	.517	1.066		
Corporate:	Status	.980	5.734*	3.604* (2,129 <i>DF</i>)	63.2
	Hours read	.457	1.454		
Professional:	Papers	.543	15.511***	7.882*** (3,128 <i>DF</i>)	87.8
	Education	.484	5.213**		
	Hours read	.314	2.218		
Operational:	<i>F</i> level insufficient for computation				

* $p < .05$

** $p < .01$

*** $p < .001$

TABLE 5
Methods of Establishing External Contacts

<i>A. Contacts Within the Organization</i>				
<i>Domains</i>	<i>Formal Channels</i>	<i>Worked Together</i>	<i>Other^a</i>	<i>Total</i>
Laboratory	150	341	64	555
Corporate	143	108	29	280
$X^2 = 49.20$ $p < .001$				
<i>B. Contacts Outside the Organization</i>				
<i>Domains</i>	<i>Formal Channels</i>	<i>Worked Together</i>	<i>Other</i>	<i>Total</i>
Professional	28	20	10	58
Operational	77	11	7	95
$X^2 = 18.6$ $p < .001$				

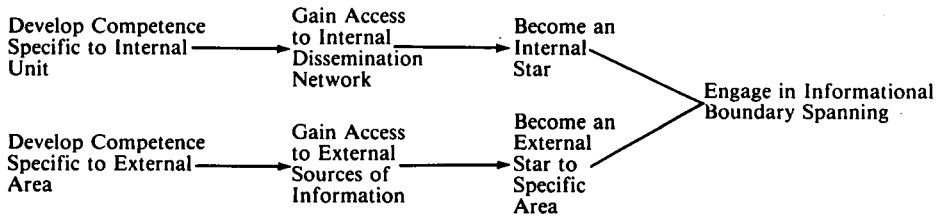
^aSee methods section of paper for detail.

DISCUSSION AND IMPLICATIONS

This research has investigated two interrelated questions: how do new ideas and information enter organizations, and what are the antecedents of those individuals who provide this informational link? The literature on boundary spanning suggests that external communication stars, internal stars, and those who are strongly linked internally and externally are each potential linking mechanisms for their more locally oriented colleagues. This research has hypothesized that due to differentiation and the concomitant specialization in languages and cognitive orientations, the transfer of information must occur in a multistep process. Informational boundary spanners must be able to translate across communication boundaries and be aware of contextual information on both sides of the boundary. It was hypothesized, therefore, that informational boundary spanning would be accomplished only by those individuals who were strongly linked internally *and* externally. The data strongly supported this hypothesis. Neither internal stars nor external stars were seen as valuable sources of new information and ideas. Extensive internal and external communication are each necessary but not sufficient conditions for informational boundary spanning. Boundary spanning individuals must gather and disseminate information and, as such, must be well linked internally *and* externally (Figure 1). Further, information boundary spanning was not limited to the hierarchy. A full 68 percent of the boundary spanning individuals were bench-level scientists or engineers.

These results strongly suggest distinguishing between informational and representational boundary spanning roles (Aldrich & Herker, 1977). This research supports other work that indicates that informational boundary spanning is a multistep process (Allen & Cohen, 1969; Wilensky, 1967).

FIGURE 1
Schematic of Informational Boundary
Spanning Model



Informational boundary spanning frequently is an informal phenomenon and is associated with job satisfaction, morale, and promotion (Allen, 1970; Blau, 1963; Keller & Holland, 1975). Furthermore, because informational boundary spanners mediate a critical resource for the organization they will have substantial informal power and status (Pettigrew, 1972; March & Simon, 1958).

Representational roles, however, focus on resource acquisition and disposal—for example, salesmen, buyers, clerks (Adams, 1976). Because such roles are evaluated, rewarded, and trained for specific purposes, their sensitivity to exceptions, opportunities, or new ideas from outside the organization is reduced. Because of such formal roles, even if a salesperson (for instance) acquires an important technical idea, the organizational impediments to the successful transfer of this information are substantial (Von Hippel, 1976; Mansfield & Wagner, 1976). Thus, although these individuals in representational roles have substantial external communication (they transmit substantial information), they may not serve as an information linking mechanism for areas within the organization. Finally, although representational roles are formally defined, these individuals face multiple and often conflicting constituencies within and outside an organization (Miles & Perreault, 1976; Kahn, Wolfe, Quinn, Snoek, & Rosenthal, 1964). Unlike informational boundary spanners, representational roles usually do not mediate critical resources for the organization and, therefore, are not an informally powerful role.

Much of the ambiguity in the literature regarding boundary spanning and its consequences may be due to confounding informational and representational communication roles. Informational roles acquire information from external areas and transmit this information internally; representational roles perform a more routine transacting/representation role. Informational roles are involved in a two-step information flow; representational roles are involved more in a one-step flow. The antecedents and consequences of these roles are different due to their contrasting functions for their peers as well as for their organizations. Future research, therefore, must distinguish between boundary spanning individuals and those individuals with high boundary spanning activity.

What are the antecedents of boundary spanning roles? These results indicate that perceived work-related competence is a basic determinant of informational boundary spanning. Perceived competence is a more powerful determinant of internal consultation than is formal status. Technical competence and status are related, but this research and that of Blau (1963), Allen (1977), and Tushman and Scanlan (1981) suggest that competence is associated with colleague consultation and may, in turn, be associated with a promotion. Formal status then further facilitates an individual's ability to serve as a boundary spanning individual. Individuals consult those whom they see as competent, independent of formal status. These internal communication stars will therefore be able to disseminate effectively information and ideas.

To provide external linkage, internal communication stars must also be well connected to external areas. Communication with professional colleagues (either within the laboratory or outside the organization) requires the recoding and synthesis of equivocal information. Communication with organizational or operational areas involves sending and/or receiving less equivocal information. Consistent with differences in the nature of information requirements to different domains, external communication stars had specialized characteristics and used distinct methods to forge links with these different areas. Communication stars to other laboratory areas were established informally (usually by working together); communication stars to professional areas outside the organization tended to be professionally oriented and relied on formal and informal mechanisms to establish links with external colleagues. Communication to organizational and operational areas, on the other hand, was more formally constrained. Corporate communication stars tended to be supervisors whose organizational linkages were formed through formal channels. Operational communication links also were established through formal channels. These results, along with those reported by Keller, Szilagyi, and Holland (1976), Tushman and Scanlan (1981), and Whitley and Frost (1973), indicate that communication stars to different external areas have specialized characteristics and use distinct methods to develop effective linkages with their respective external areas.

External communication stars were not seen as more competent than nonstars, they were not seen as valuable sources of new information, and they were not strongly linked internally (only 25 percent were also internal communication stars). Internal communication stars, however, were seen as technically competent independent of their external linkage. This perceived local task expertise may be associated with informal power; internal communication stars who were not externally linked may well be influential for administrative or organizational decision making. Boundary spanning individuals with their internal and external expertise may be influential in both technical as well as administrative decision making. Those individuals who are simply external communication stars most likely will not be influential in either technical or administrative decision making. Future

research must investigate the roles of internal stars and boundary spanning individuals in organizational decision making (Pettigrew, 1972; Spekman, 1979; Aldrich & Herker, 1977).

How, then, does the project leader or technical staff member (discussed earlier) acquire relevant, timely, and useful external information? This research indicates that individuals approach informal boundary spanning individuals who steer them to the relevant information or provide that information directly. This is an informal process in which individuals only approach those whom they see as technically competent and who have well developed linkages to specific external areas. Different individuals evolve to handle distinct external information areas (Tushman & Scanlan, 1981).

Can this emergent phenomenon be managed? This research indicates that management can facilitate the evolution and development of boundary spanning individuals (Farris, 1972; Allen, 1977). A subset of an organization's technically competent staff can be selected (Keller & Holland, 1978) and given the opportunity to develop contacts and broaden their exposure with other areas of the corporation through transfers, work assignments, and in-house training. Management could encourage more extensive external communication through work assignments, travel allowances, or through special education programs. Managers, then, can select potential boundary spanning individuals and systematically facilitate the development of their internal and external communication networks. Finally, although formalizing informational boundary roles might be dysfunctional (Miles & Perreault, 1976; Keller et al., 1976), their promotion may be functional. Promotion of boundary spanning individuals not only recognizes and rewards that activity, but also permits those individuals to develop more diverse internal and external communication networks (Tushman & Scanlan, 1981).

Beyond these formal mechanisms, management also must create the climate in which it is legitimate to ask for assistance, in which providing this expertise is recognized, valued, and rewarded. Norms must evolve so that those with the relevant task expertise, whether they be supervisors or peers, are encouraged to provide work-related assistance to their colleagues. This emergent boundary spanning process can then be facilitated through formal and informal mechanisms.

In summary, this research has investigated one mechanism by which information is imported into organizations. Building on existing research, these results indicate that informational boundary spanning occurs in a two-step process by individuals who are able to gather information from external areas and disseminate that information to their colleagues. These key individuals are seen as among the most competent individuals in their unit, and they have specialized characteristics to facilitate communication with distinct external areas. Future research must clearly distinguish between informational (a two-step process) versus representational (a single step process) boundary spanning roles, investigate other mechanisms for information transfer, and explore the role of boundary spanning individuals in organizational decision making.

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